1. dig

Queries DNS records for a domain.

```
root@Uday:~# dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.8-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 18782
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             247     IN      A       142.250.77.142

;; Query time: 11 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Mon Nov 02 19:47:56 UTC 2026
;; MSG SIZE  rcvd: 55
```

2. namei

Displays every directory component in a file path along with its permissions.

```
root@Uday:~# namei -l /var/www/html/index.html
f: /var/www/html/index.html
drwxr-xr-x root root /
drwxr-xr-x root root var
                     www - No such file or directory
root@Uday:~#
```

3. install

Creates directories or installs files with specified permissions.

```
root@Uday:~# sudo install -d -m 755 /var/www/myapp
root@Uday:~# ls /var/www
myapp
root@Uday:~#
```

4. fallocate

Creates a file with a specified size without writing all the contents manually.

```
vanam@Uday:~$ fallocate -l 100M testfile
vanam@Uday:~$ ls -l
total 102468
drwxr-xr-x 6 vanam vanam       4096 Nov  7  2025 FeedbackApp
-rw-r--r-- 1 vanam vanam          6 Oct 17 19:46 chunk_00
-rw-r--r-- 1 vanam vanam          6 Oct 17 19:46 chunk_01
-rw-r--r-- 1 vanam vanam          6 Oct 17 19:46 chunk_02
-rw-r--r-- 1 vanam vanam          3 Oct 17 19:46 chunk_03
drwxr-xr-x 2 vanam vanam       4096 Oct 12 09:53 devops
-rw-r--r-- 1 vanam vanam         16 Oct 17 19:46 file1
-rw-r--r-- 1 vanam vanam         14 Oct 13 09:11 file1.txt
-rw-r--r-- 1 vanam vanam         18 Oct 17 19:46 file2
drwxr-xr-x 2 vanam docker      4096 Nov  7  2025 jenkins-pytho
drwxr-xr-x 2 vanam vanam       4096 Oct 14 10:52 linux-day4
drwxr-xr-x 2 vanam vanam       4096 Oct 12 10:16 member2_lab
-rw-r--r-- 1 vanam vanam         14 Oct 17 19:46 names.txt
-rw-r--r-- 1 vanam vanam         21 Oct 17 19:46 numbers.txt
drwxr-xr-x 2 vanam vanam       4096 Oct 13 09:12 practice
-rw-r--r-- 1 vanam vanam         16 Oct 17 19:46 skills.txt
-rw-r--r-- 1 vanam vanam  104857600 Nov  2 19:54 testfile
drwxr-xr-x 2 vanam vanam       4096 Oct 11 11:15 week0
```

- `-l` – Specifies file size.
- `100M` – Creates a 100 MB file.

5. sha256sum

Calculates a SHA-256 checksum to verify file integrity.

```
root@Uday:~# sha256sum index.html
2e7382e12c44aec08ff9144c6d51fe94172fba44d2805ef194fbacaca94a695a  index.html
root@Uday:~#
```

6. systemctl

Checks detailed service status.

```
root@Uday:~# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-11-02 19:17:56 UTC; 32min ago
   Main PID: 179 (java)
      Tasks: 53 (limit: 9330)
     Memory: 559.1M (peak: 633.6M)
        CPU: 37.728s
     CGroup: /system.slice/jenkins.service
             └─179 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins>

Nov 02 19:17:51 Uday jenkins[179]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Nov 02 19:17:51 Uday jenkins[179]: [LF]>
Nov 02 19:17:51 Uday jenkins[179]: [LF]> ************************************************************
Nov 02 19:17:51 Uday jenkins[179]: [LF]> ************************************************************
Nov 02 19:17:51 Uday jenkins[179]: [LF]> ************************************************************
Nov 02 19:17:56 Uday jenkins[179]: 2026-11-02 19:17:56.831+0000 [id=65]        INFO        jenkins.InitReactorRunner$1#>
Nov 02 19:17:56 Uday jenkins[179]: 2026-11-02 19:17:56.875+0000 [id=40]        INFO        hudson.lifecycle.Lifecycle#o>
Nov 02 19:17:56 Uday systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Nov 02 19:17:58 Uday jenkins[179]: 2026-11-02 19:17:58.579+0000 [id=88]        INFO        h.m.DownloadService$Download>
Nov 02 19:17:58 Uday jenkins[179]: 2026-11-02 19:17:58.580+0000 [id=88]        INFO        hudson.util.Retrier#start: P>
lines 1-20/20 (END)
```

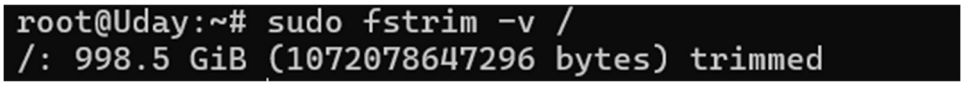
7. logrotate

Manages log rotation and prevents log files from consuming excessive disk space.



8. fstrim

Releases unused filesystem blocks to storage devices that support discard/TRIM.



9. at

Schedules a command to execute once at a specified time.



10. crontab

Schedules recurring jobs.

